

$$\{x^\mu, \chi_A^\mu\}, \mu=0, \dots, D-1$$

$$x^\mu \in \mathbb{M}^D$$

$$\chi_A^\mu: \mathbb{M}^D \otimes \mathbb{C}^2 \quad A=1, 2$$

Grassmann odd, two-dim Maj. spinors

- χ_A^μ is not a spinor of the background spacetime but is a D-dim vector

- in $\mathbb{M}^D$: χ_A^μ is a D-dim vector

- on Σ^2 : χ_A^μ is a 2d-spinor in $\text{Spin}(2)$

→ These theories possess (1,1) supersymmetry on Σ_2 with Aux. fields N^μ .

- Chiral Supermultiplet $(x^\mu, \chi_A^\mu, N^\mu)$

$$\int d^2\zeta \left(-\frac{1}{2} \partial_\alpha x^\mu \bar{\partial} \chi^\alpha - \frac{i}{2} \bar{\chi}^\mu \not{\partial} \chi^\nu + \frac{1}{2} N^\mu N^\nu \right) \Omega_{\mu\nu}$$

→ Couple this theory to (1,1) SUGRA

$$S = \frac{1}{2\pi\alpha'} \int d^2\zeta e \left\{ -\frac{1}{2} e_\alpha^\mu \dots \right\} = \int d^2\zeta \mathcal{L}$$

$$e \equiv \det(e_\alpha^\mu), \quad \not{\partial} = \gamma^\alpha e_\alpha^\mu \partial_\mu$$

$$M: \mathbb{M}^D$$

$$\alpha: \mathbb{M}^2$$

$$a: \Sigma^2$$